

A New Knowledge Mining and Root Cause Analysis Methodology for Multivariate Time Series

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Abstract—Root cause analysis (RCA) aims to discover the root causes of abnormal events. Causal relations reveal the evolution process of abnormal events, which plays a crucial role in RCA. However, existing methods neither explicitly emphasize the “AND/OR” relations among causes, nor consider the synergy effects owned by non-causal variables on causal rules, thereby affecting the credibility of RCA. To address the issues, by fusing Petri nets and Bayesian networks, this study proposes a new knowledge mining and RCA methodology for multivariate time series, called synergy-incorporated Bayesian time Petri net. It integrates the advantages of Petri nets in modeling and analyzing complex temporal dependencies and Bayesian networks in evidence reasoning. It takes into account “AND/OR” relations and synergy effects in temporal knowledge mining and RCA. Two cases are employed to verify its performance in knowledge mining and RCA, including a case study of quality anomaly detection of solar panels and the Tennessee-Eastman process. The experimental results from both cases indicate that it can effectively consider “AND/OR” relations and synergy effects. When applied to the former, it outperforms the state-of-the-art RCA methods in accuracy by over 11%.

Index Terms—Bayesian network (BN), knowledge mining, multivariate time series, Petri net (PN), root cause analysis (RCA), synergy effect.

I. INTRODUCTION

ROOT cause analysis (RCA) focuses on clarifying the propagation path of abnormal events and locating their

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root causes. It is crucial for ensuring the stability of production systems and improving product quality. Existing RCA methods are multivariate statistical analysis-based [1]–[3], machine learning-based [4]–[7], and causality-driven [8]–[10].

Multivariate statistical analysis-based RCA methods employ correlation analysis, including principal component analysis [1], partial least squares [2], and independent component analysis [3], to reveal the statistical characteristics of fault data. This allows for the identification of the main factors contributing to the fault. While these approaches can be used to analyze complex interactions among multiple variables, they are often restricted by specific model assumptions (e.g., linear relationships or normal distributions) and are highly sensitive to data quality.

Machine learning-based RCA methods utilize machine learning algorithms to automatically learn anomaly patterns from data. These methods can be categorized into unsupervised [4], [5] and supervised learning-driven approaches [6], [7]. The former, e.g., association rule mining [4] and k -means [5], are capable of automatically discovering hidden anomaly patterns from unlabeled data, thereby identifying the crucial factors contributing to the problems. The latter, e.g., decision trees [6] and neural networks [7], transform an RCA problem into a classification or regression problem. They are trained using labeled historical anomaly data to learn the mapping relationship between input features and labels, enabling them to predict potential root causes based on anomaly patterns. Deep neural networks, such as long short-term memory [7], recurrent neural networks [11], convolutional neural networks [12], and graph neural networks [13], can handle complex data structures and nonlinear features, making them suitable for RCA in high-dimensional data scenarios. The work [7] proposes an attention-based long short-term memory approach for alarm root cause diagnosis in chemical processes, which can effectively identify causal relations with long time delays. The work [11] proposes a root cause diagnosis framework for quality-related faults in manufacturing processes. It utilizes a gated recurrent neural network to capture nonlinear dynamic causal relations, thereby identifying fault propagation paths and locating the root causes of faults. The work [12] uses convolutional neural networks to extract defect layout features from large-scale integrated circuits, in order to analyze the root causes of the layouts generating defects. It is observed that machine learning-based RCA methods are prone to over-

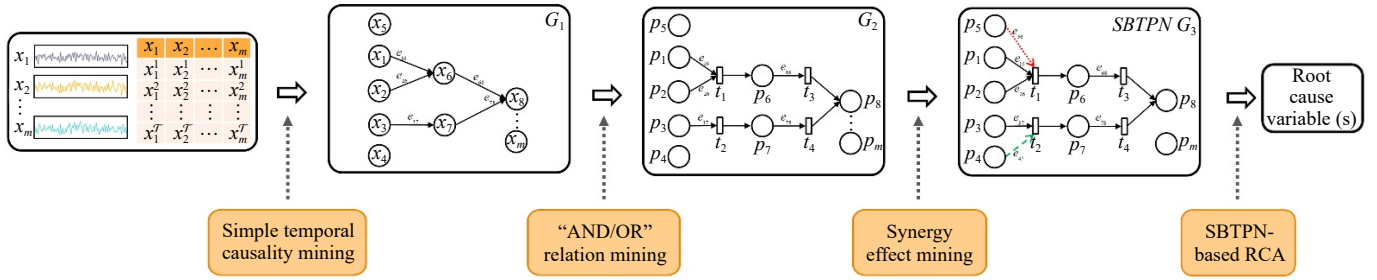


Fig. 1. The architecture of the proposed methodology.

fitting, and their results generally offer poor interpretability [14], [15].

By revealing the underlying causal mechanisms among system variables, causality-driven RCA methods can precisely identify the root causes of abnormal events. For complex industrial processes, their evolutionary states are usually recorded in the form of multivariate time series (MTS) [16] that depict the states of multiple variables changing over time. Thus, mining causal relations from MTS is crucial for RCA of such processes. Existing causal discovery methods for MTS include constraint-based [17], [18], score-based [8], [19], structural equation-based [20], [21], Granger causality-based [9], [22], and deep learning-based ones [10], [23]. Specifically, constraint-based ones adopt conditional independence tests to determine the causal relation between two observed variables. Score-based ones discover an optimal directed acyclic graph (DAG) from candidate DAGs by using a score function that measures their fitness to the data. A structural equation-based one is a multiple-equation regression model, where each variable is explained by a regression equation consisting of its direct causes and additional noises. A Granger causality-based one is a predictability-based temporal causal discovery method. It quantifies the extent to which the past of a time series helps predict the future evolution of another time series. Deep learning-based ones utilize the advantage of deep neural networks in establishing complex nonlinear relationships and long-term dependencies to discover causal relations from nonlinear and nonstationary process data. The key to achieving high-quality RCA lies in the precise modeling of the discovered causal relations. Classical causal modeling paradigms include Bayesian networks (BNs) [24]–[26], Petri nets (PNs) [27]–[29], and fault trees [30], [31]. Among them, BNs and PNs are two important complex knowledge modeling methodologies. A BN is a directed acyclic graph with strong uncertainty reasoning ability. It adopts an evidence propagation mechanism for RCA. PNs are a mathematical analysis tool with a graphical representation. They are employed to model and analyze complex dependencies. The extended PNs presented in [32] and [33] support fuzzy backward reasoning for RCA.

There may be “AND/OR” relations among causes as well as synergy effects of non-causal variables on causal rules. An “AND” relation represents a conjunctive structure in which a causal rule holds if and only if all of its causes are true. An “OR” relation represents a disjunctive structure in which a causal rule holds if at least one of its causes is true. If a non-causal variable can increase or decrease the probability that a

causal rule holds, it has a catalytic or inhibitory effect on the rule. In this paper, both catalytic and inhibitory effects are collectively referred to as synergy effects.

“AND/OR” relations and synergy effects play an important role in RCA. They precisely depict the complex dependencies among the factors contributing to an abnormal event. Discovering such relations and effects from MTS is crucial for constructing a fine-grained causal model and identifying the root causes of abnormal events based on the model. However, existing temporal causal discovery methods neither explicitly emphasize “AND/OR” relations nor consider synergy effects, which affects the credibility of RCA.

To address the issues, by fusing PNs and BNs, this study proposes a new knowledge mining and RCA methodology for MTS, called synergy-incorporated Bayesian time Petri net (SBTPN). It considers “AND/OR” relations and synergy effects in temporal knowledge mining and RCA. As shown in Fig. 1, its basic idea is as follows: 1) learning the simple temporal causal relations from MTS; 2) mining the “AND/OR” relations among causes based on the temporal causal relations; 3) discovering the synergy effects of non-causal variables on causal rules based on the “AND/OR” relations; and 4) performing RCA based on the “AND/OR” relations and synergy effects.

This work aims to make the following new contributions:

1) It proposes a framework for mining complex temporal causal relations by integrating a one-dimensional dilated convolutional neural network with conditional mutual information and conditional probability. Compared to existing temporal causal discovery methods [9], [10], [34], the proposed framework can discover finer-grained temporal causal relations from MTS.

2) It introduces the concept of “AND/OR” relations and synergy effects into RCA by fusing PNs and BNs. It proposes a new RCA method that takes into account synergy effects, which is not seen in any existing RCA methods. The proposed RCA method has lower data quality requirements and provides more explainable and finer-grained RCA results than existing RCA methods [23], [35], [36], thereby enabling the development of more targeted preventive measures for abnormal events.

The rest of this paper is organized as follows. Section II discusses the challenges faced by existing temporal causal discovery methods. Section III introduces the preliminaries of PNs and BNs. Section IV presents the formal definition of SBTPN. Section V proposes the method for mining “AND/OR” relations and synergy effects from MTS, and

variable $x_i \in X$. The joint probability distribution over X is determined as

$$\Pr(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \Pr(x_i | P_A(x_i)). \quad (1)$$

BN does not explicitly emphasize time-constrained “AND/OR” relations and synergy effects, which affects the credibility of its RCA results. To address this issue, this study proposes an RCA method that considers these factors and is detailed in Section VI. Prior to this, it is necessary to model the time-constrained “AND/OR” relations and synergy effects. Next, a temporal causal modeling paradigm considering them is presented by fusing PNs and BNs.

IV. SYNERGY-INCORPORATED BAYESIAN TIME PETRI NET (SBTPN)

This section presents the formal definition of SBTPN.

Definition 1: An SBTPN is a 9-tuple $SBTPN = (X, P, \beta, T, F, S, E, \Pr, M)$, where

- 1) $X = \{x_1, x_2, \dots, x_m\}$ is a finite set of variables.
- 2) $P = \{p_1, p_2, \dots, p_m\}$ is a finite set of places.
- 3) $\beta: P \rightarrow X$ is an association function defining a one-to-one mapping between a variable in X and a place in P .
- 4) $T = \{t_1, t_2, \dots, t_n\}$ is a finite set of transitions satisfying $P \cap T = \emptyset$.
- 5) $F \subseteq (P \times T) \cup (T \times P)$ is a set of flow relations defining solid arcs from places to transitions or from transitions to places.
- 6) $S \subseteq P \times T$ is a set of synergy relations defining dashed/dotted arcs from places to transitions.
- 7) $E: P \times P \rightarrow [0, +\infty)$ is a time delay function, where $e_{ij} = E(p_i, p_j)$ ($e_{ij} \in [0, +\infty)$; $i, j \in \{1, 2, \dots, m\}$) describes the time delay of the impact of variable $\beta(p_i)$ on variable $\beta(p_j)$.
- 8) $\Pr: P \cup T \rightarrow [0, 1]$ is a probability function mapping each place and transition to a probability value in $[0, 1]$.
- 9) $M = \bigcup_{i=1}^m \{m(p_i)\}$ is a marking encoding the probability distribution of each place in P .

Note that: 1) The pre-set of a node $v \in P \cup T$ is defined as $\bullet v = \{u \in P \cup T | (u, v) \in F\}$ and its post-set as $v^\bullet = \{u \in P \cup T | (v, u) \in F\}$; 2) The set of places that have synergy effects on transition t is defined as ${}^\circ t = \{p | (p, t) \in S \wedge (p, t) \notin F\}$. The set of transitions affected by the synergy effects of place p is defined as $p^\circ = \{t | (p, t) \in S \wedge (p, t) \notin F\}$. We assume that $|p_i^\bullet \cap {}^\bullet p_j| = 1$ ($|p_i^\circ \cap {}^\circ p_j| = 1$) for each effect variable $\beta(p_j)$ with its direct cause (synergy factor) $\beta(p_i)$. If $\beta(p_i)$ is the direct cause of $\beta(p_j)$, $E(p_i, p_j)$ is associated with the solid arc from p_i to $t_j = p_i^\bullet \cap {}^\bullet p_j$. If $\beta(p_i)$ is the catalytic (inhibitory) factor for producing $\beta(p_j)$, $E(p_i, p_j)$ is associated with the dashed (dotted) arc from p_i to $t_j = p_i^\circ \cap {}^\circ p_j$; 3) Throughout this paper, $\hat{p}$ ($\check{p}$) represents that $\beta(p)$ is present (absent), and $\hat{t}$ ($\check{t}$) indicates that t is enabled (disabled); and 4) The initial marking of an SBTPN is determined by the prior probabilities of the place-associated variables, which can be obtained by calculating the frequencies of these variables in the sample data. In data-scarce scenarios, they can be determined by expert knowledge.

Consider SBTPN in Fig. 2. We have $X = \{x_1, x_2, \dots, x_9\}$, $P = \{p_1, p_2, \dots, p_9\}$, $x_i = \beta(p_i)$ ($i = 1, 2, \dots, 9$), $T = \{t_1, t_2, t_3, t_4\}$, $\bullet p_3 = \emptyset$, $p_3^\bullet = \{t_2\}$, $\bullet t_2 = \{p_3\}$, $t_2^\bullet = \{p_5\}$, $p_8^\circ = t_2$, ${}^\circ t_2 = \{p_8\}$, $e_{35} = E(p_3, p_5) = 3$, $e_{85} = E(p_8, p_5) = 2$, and $m(p_3) = \Pr(\hat{p}_3)$.

The joint probability distribution of SBTPN is defined as

$$\Pr(r_1, r_2, \dots, r_{m+n}) = \prod_{i=1}^{m+n} \Pr(r_i | {}^\bullet r_i) \quad (2)$$

where r_i ($i = 1, 2, \dots, m+n$) denotes a place or transition and $m+n$ represents the total number of places and transitions. To calculate joint probability distribution, one needs to determine the conditional probability of a transition given its input places and that of a place given its input transitions, to be computed in Section V-E.

The SBTPN has two key characteristics: the integration of PN transitions into the probability distribution of BNs, and the introduction of synergy effects into BNs. From the perspective of knowledge modeling and RCA, their appropriateness and necessity are as follows. By incorporating PN concepts into BN, the complexity of constructing BN can be reduced, thereby improving the efficiency of BN inference. By introducing synergy effects into BN, the catalytic/inhibitory effects objectively existing in the real world can be formally modeled, thereby improving the reasonableness of causality-driven RCA. For a more detailed discussion of these two characteristics, please refer to the [Supplementary File](#).

V. MINING OF SBTPN FROM MTS

In this paper, the temporal causal relations without considering “AND/OR” relations and synergy effects are called simple ones. Those considering them are called complex ones. Existing temporal causal discovery methods focus on learning simple temporal causal relations. They do not explicitly emphasize the “AND/OR” relations and synergy effects constrained by time. To address this issue, this paper proposes a method for mining SBTPN from MTS. It consists of the following steps: 1) learning the simple temporal causal relations from MTS; 2) mining the “AND/OR” relations constrained by time based on the discovered causal relations; and 3) discovering the synergy effects constrained by time based on the identified “AND/OR” relations.

A. Learning Simple Temporal Causal Relations

The temporal causal discovery framework (TCDF) proposed in [39], which is a 1D dilated convolutional neural network with a causal validation step, is employed to learn the simple temporal causal relations from MTS. Its structure is called attention-based dilated depthwise separable temporal convolutional network (AD-DSTCN). Each time series x_j ($j = 1, 2, \dots, m$) is associated with an AD-DSTCN j , which is composed of m dilated convolutional neural networks (DCNNs) with 1D kernels. The input of DCNN $i \in \{1, 2, \dots, m\}$ in AD-DSTCN j is a time series x_i and its outputs are the predictive value $\hat{x}_j$, the attention score a_{ij} that is used to determine whether x_i is the cause of x_j , and the kernel weight w_{ij} that is used to calculate the time delay of the impact of x_i on x_j . The larger the value of a_{ij} , the higher the

correlation between x_i and x_j . However, the correlation does not equal causality. To determine whether x_i is the cause of x_j , an intervention dataset with respect to x_i can be obtained by randomly permuting the value of x_i in the original input dataset while the values of the other time series in it remain unchanged. If x_i is the cause of x_j , the loss value of predicting x_j on the intervention dataset with respect to x_i should be significantly higher than that on the original input dataset. In addition, $w_{ij} = [w_{ij}^{\tau-R+1}, w_{ij}^{\tau-R+2}, \dots, w_{ij}^{\tau}]$, where R is the number of time steps seen by the sliding kernel and $w_{ij}^{\tau-y}$ ($y = 0, 1, \dots, R-1$) represents the importance of $x_i^{\tau-y}$ in predicting x_j^{τ} .

Definition 2: For two time series x_i and x_j , if $w_{ij}^{\tau-k} = \max\{w_{ij}^{\tau-R+1}, w_{ij}^{\tau-R+2}, \dots, w_{ij}^{\tau}\}$ ($k \in \{0, 1, \dots, R-1\}$), the time delay of the impact of x_i on x_j is k .

TCDF outputs a causal graph that depicts the simple temporal causal relations among all time series. More details about simple temporal causality mining are discussed in the [Supplementary File](#).

B. Quantifying Impact of Input Variable on Target One

The total number of time steps of a time series is denoted as ζ . If the time delay of time series x_i affecting time series x_j is e_{ij} ($e_{ij} \in \mathbb{N}$), x_j^{τ} ($\tau = 1, 2, \dots, \zeta$) is affected by $x_i^{\tau-e_{ij}}$, $x_i^{\tau-e_{ij}+1}$, $\dots$, and x_i^{τ} , and the corresponding impacts can be quantified as TCDF's kernel weights $w_{ij}^{\tau-e_{ij}}$, $w_{ij}^{\tau-e_{ij}+1}$, $\dots$, and w_{ij}^{τ} , respectively.

Definition 3: $\mu_{ij} = \bigcup_{\tau=1}^{\zeta} \{\mu_{ij}^{\tau}\}$ is a dependence strength index denoting the impact of x_i on x_j within ζ time steps, where μ_{ij}^{τ} represents the impact of x_i on x_j at time step τ , i.e.,

$$\mu_{ij}^{\tau} = \frac{\sum_{l=\tau-e_{ij}}^{\tau} x_i^l \times w_{ij}^l}{\sum_{l=\tau-e_{ij}}^{\tau} w_{ij}^l} \quad (3)$$

where μ_{ij} is the foundation for mining complex temporal causal relations and performing RCA based on these relations.

C. Mining "AND/OR" Relations Constrained by Time

Given a causal graph G_1 learned by TCDF, let $Parents(G_1, x_k)$ denote the set of causes of time series x_k in G_1 . Any time series x_i and x_j belonging to $Parents(G_1, x_k)$ affect the value of x_k . By Definition 3, their corresponding impacts can be quantified as μ_{ik} and μ_{jk} , respectively. If there is an "AND" relation between x_i and x_j pointing to x_k , the value of x_k is true only if the values of μ_{ik} and μ_{jk} are both true. However, due to the influence of noise, it becomes difficult to accurately identify such "AND" relation by directly observing the values of μ_{ik} , μ_{jk} , and x_k . This problem can be alleviated by introducing conditional mutual information when determining "AND" relations. If the value of mutual information between μ_{ik} (μ_{jk}) and x_k when μ_{jk} (μ_{ik}) is true is greater than that when μ_{jk} (μ_{ik}) is false, there may be an "AND" relation between x_i and x_j pointing to x_k . The idea for mining the "AND" relations constrained by time is quantified as

$$\begin{cases} \Phi(x_i, x_j, x_k) = \frac{1}{1 + \exp(\phi_1 - \phi_2)} \\ \phi_1 = I(\mu_{ik}; x_k | \hat{\mu}_{jk}) \\ \phi_2 = I(\mu_{jk}; x_k | \hat{\mu}_{ik}) \end{cases} \quad (4)$$

where ϕ_1 (ϕ_2) represents the mutual information between μ_{ik} and x_k when μ_{jk} is false (true). In this paper, variable v is considered to be true only if $v \in [0.5, 1]$. $\Phi(x_i, x_j, x_k)$ is used to determine the strength of the "AND" relation between x_i and x_j pointing to x_k . This work supposes that there is an "AND" relation between x_i and x_j pointing to x_k only if $\Phi(x_i, x_j, x_k) > 0.5$ and $\Phi(x_j, x_i, x_k) > 0.5$. The detailed algorithm for discovering "AND/OR" relations from MTS can be found in the [Supplementary File](#).

D. Discovering Synergy Effects Constrained by Time

Let $\mathbb{N}_h = \{1, 2, \dots, h\}$. Consider transition t_k with $\bullet t_k = \bigcup_{a \in \mathbb{N}_h} \{p_a\}$ and $t_k^{\bullet} = p_j$. $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\})$ represents the probability of variable $\beta(p_j)$ being true under the condition that all its causes have an impact on it. Given a synergy place p_x with respect to t_k , $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{xj})$ represents the probability of variable $\beta(p_j)$ being true under the condition that all its causes and synergy factor $\beta(p_x)$ have an impact on it. If p_x has a catalytic effect on t_k , we have $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{xj}) > \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\})$. Similarly, if p_x has an inhibitory effect on t_k , we have $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{xj}) < \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\})$. The method for mining the synergy effects constrained by time can be quantified as

$$\begin{cases} \Psi(\bullet t_k; t_k^{\bullet}; p_x) = \frac{1}{1 + \exp(\psi_1 - \psi_2)} \\ \Upsilon(\bullet t_k; t_k^{\bullet}; p_x) = \frac{1}{1 + \exp(\psi_2 - \psi_1)} \\ \bullet t_k = \bigcup_{a \in \mathbb{N}_h} \{p_a\} \\ t_k^{\bullet} = p_j \\ \psi_1 = \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}) \\ \psi_2 = \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{xj}) \end{cases} \quad (5)$$

where $\Psi(\bullet t_k; t_k^{\bullet}; p_x)$ and $\Upsilon(\bullet t_k; t_k^{\bullet}; p_x)$ are used to depict the strength of the catalytic effect of p_x on t_k and that of the inhibitory effect of p_x on t_k , respectively. Let η_1 and η_2 denote the thresholds for determining a catalytic effect and an inhibitory one, respectively, which satisfy $\eta_1, \eta_2 \in [0.5, 1]$. This paper supposes that: 1) If $\Psi(\bullet t_k; t_k^{\bullet}; p_x) > \eta_1$, p_x has a catalytic effect on t_k ; and 2) If $\Upsilon(\bullet t_k; t_k^{\bullet}; p_x) > \eta_2$, p_x has an inhibitory effect on t_k . The detailed algorithm for mining synergy effects from MTS is referred to the [Supplementary File](#).

In fact, noise can interfere with the discovery of "AND/OR" relations and synergy effects. A detailed discussion is provided in the [Supplementary File](#).

E. Conditional Probability Distribution of SBTPN

For a transition t_k with $\bullet t_k = \bigcup_{a \in \mathbb{N}_h} \{p_a\}$ and $t_k^{\bullet} = p_j$, t_k is enabled only if all the variables $\bigcup_{a \in \mathbb{N}_h} \{\beta(p_a)\}$ have an impact on variable $\beta(p_j)$. Thus, let $\Pr(\hat{t}_k | \hat{p}_1, \hat{p}_2, \dots, \hat{p}_h) = \Pr(\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\})$ and $\Pr(\hat{t}_k | \hat{p}_1, \hat{p}_2, \dots, \hat{p}_h) = \Pr(\hat{t}_k | \hat{p}_1, \hat{p}_2, \dots, \hat{p}_h) = \dots = \Pr(\hat{t}_k | \hat{p}_1, \hat{p}_2, \dots, \hat{p}_h) = 0$. $\beta(p_j)$ is true only if t_k

is enabled. Thus, let $\Pr(\hat{p}_j|\hat{t}_k) = \Pr(\hat{p}_j|\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\})$ and $\Pr(\hat{p}_j|\hat{t}_k) = 0$.

VI. SBTPN-BASED RCA

Existing RCA methods [10], [22], [23] are unable to characterize the synergy effects owned by non-causal variables on causal rules, and many of them fail to consider the “AND” relations among causes, which affects the credibility of RCA. To address the issue, a new RCA method considering “AND/OR” relations and synergy effects is proposed based on the SBTPN, as shown next.

A. RCA Considering “AND/OR” Relations and Synergy Effects

Let $CauseSet(p_j)$ denote the places associated with the direct or indirect causes of variable $\beta(p_j)$, and $\hat{p}_j$ represent that $\beta(p_j)$ is in an abnormal state. Any variable $\beta(p_i)$ satisfying $p_i \in CauseSet(p_j)$ is the potential root cause of the abnormality of $\beta(p_j)$. $\Pr(\hat{p}_i|\hat{p}_j)$ depicts the possibility that $\hat{p}_i$ is the root cause of $\hat{p}_j$. The larger the value of $\Pr(\hat{p}_i|\hat{p}_j)$, the greater the possibility that $\beta(p_i)$ is the root cause of the abnormality of $\beta(p_j)$.

$\Pr(\hat{p}_i|\hat{p}_j)$ can be affected by synergy effects. Consider a transition t_k with $\bullet t_k = \bigcup_{a \in \mathbb{N}_h} \{p_a\}$ and $t_k^\bullet = p_j$. Let $\mathbb{C}(t_k)$ ($\mathbb{I}(t_k)$) denote the set of places that have catalytic (inhibitory) effects on t_k . For $\forall p_i \in \bullet t_k$ and $\forall p_x \in \mathbb{C}(t_k)$, due to the catalytic effect of p_x on t_k , the probability value $\Pr(\hat{p}_i|\hat{p}_j)$ increases, yielding

$$\begin{aligned} \Pr(\hat{p}_i|\hat{p}_j)_{x^+} &= \Pr(\hat{p}_i|\hat{p}_j) + \lambda_x^+ \Pr(\hat{p}_x|\hat{p}_j)(1 - \Pr(\hat{p}_i|\hat{p}_j)) \\ \text{s.t. } \begin{cases} \lambda_x^+ = \frac{\psi_2 - \psi_1}{1 - \psi_1} \\ \psi_1 = \Pr(\hat{p}_j|\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}) \\ \psi_2 = \Pr(\hat{p}_j|\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{xj}) \end{cases} \end{aligned} \quad (6)$$

where $\Pr(\hat{p}_i|\hat{p}_j)_{x^+}$ denotes the probability that $\hat{p}_i$ is the root cause of $\hat{p}_j$ under the catalytic effect of p_x on t_k .

Given $p_x \in \mathbb{I}(t_k)$, under the inhibitory effect of p_y on t_k , the probability value $\Pr(\hat{p}_i|\hat{p}_j)$ decreases, yielding

$$\begin{aligned} \Pr(\hat{p}_i|\hat{p}_j)_{y^-} &= \Pr(\hat{p}_i|\hat{p}_j) - \lambda_y^- \Pr(\hat{p}_y|\hat{p}_j)\Pr(\hat{p}_i|\hat{p}_j) \\ \text{s.t. } \begin{cases} \lambda_y^- = \frac{\psi_1 - \psi_2}{\psi_1} \\ \psi_1 = \Pr(\hat{p}_j|\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}) \\ \psi_2 = \Pr(\hat{p}_j|\bigcup_{a \in \mathbb{N}_h} \{\hat{\mu}_{aj}\}, \hat{\mu}_{yj}) \end{cases} \end{aligned} \quad (7)$$

where $\Pr(\hat{p}_i|\hat{p}_j)_{y^-}$ denotes the probability that $\hat{p}_i$ is the root cause of $\hat{p}_j$ under the inhibitory effect of p_y on t_k .

Let $\Pr(\hat{p}_i|\hat{p}_j)'$ denote the probability that $\hat{p}_i$ is the root cause of $\hat{p}_j$ considering all synergy effects imposed on t_k , which can be characterized by

$$\Pr(\hat{p}_i|\hat{p}_j)' = \frac{1}{g+q} \left(\sum_{x=1}^g \Pr(\hat{p}_i|\hat{p}_j)_{x^+} + \sum_{y=1}^q \Pr(\hat{p}_i|\hat{p}_j)_{y^-} \right) \quad (8)$$

where g (q) represents the number of places that have catalytic (inhibitory) effects on t_k .

The larger the value of $\Pr(\hat{p}_i|\hat{p}_j)'$, the greater the possibility that $\beta(p_i)$ is the root cause of the abnormality of $\beta(p_j)$. How-

ever, it is difficult to accurately identify whether $\beta(p_i)$ is the root cause of $\beta(p_j)$ relying solely on the abnormal states of $\beta(p_i)$ and $\beta(p_j)$. This is because all the variables located in the causal path from $\beta(p_i)$ to $\beta(p_j)$ are involved in the propagation of the abnormal state of $\beta(p_i)$. In addition, for transition $t_k \in p_i^\bullet$ located in this causal path, all the variables directly pointing to t_k being in an abnormal state is the necessary condition for propagation of the abnormal state of $\beta(p_i)$. This work supposes that $\beta(p_i)$'s anomaly is the root cause of $\beta(p_j)$'s anomaly only if all the variables directly pointing to t_k and all the variables located in the causal path from $\beta(p_i)$ to $\beta(p_j)$ are in an abnormal state. The detailed RCA algorithm is referred to the [Supplementary File](#).

B. Computational Complexity Analysis

The computational complexity of SBTPN comes from four sources: simple temporal causality, “AND/OR” relation and synergy effect discoveries, and RCA considering “AND/OR” relations and synergy effects. Given MTS $X = \{x_1, x_2, \dots, x_m\}$ with sample size l , since one needs to identify the temporal causal relation between any two time series in X within l time steps, the time complexity of discovering simple temporal causalities is $O(m^2l)$. Because one needs to determine the “AND” relation between any two causes of variable x_i ($i = 1, 2, \dots, m$), the time complexity of mining “AND/OR” relations is $O(m^3)$. For two variables with a direct causal relation, this work supposes that their corresponding places can only be associated through one transition. Hence, the maximum number of transitions corresponding to X is $m(m-1)/2$. In the worst case, a transition can have m synergy places. Thus, the time complexity of discovering synergy effects and that of RCA based on SBTPN are both $O(m^3)$.

SBTPN extends existing temporal causal discovery methods by identifying “AND/OR” relations among causes and synergy effects of non-causal variables on causal rules. The studies [40] and [41] focus on mining “AND” relations from non-temporal observational data. The following discusses the computational complexity of SBTPN and these two studies in mining the “AND” relations that point to a single effect variable. For the causal discovery methods proposed in [40] and [41] and SBTPN, their computational complexities for mining such relations are $O(m^m)$, $O(2^m)$, and $O(m^2)$, respectively, where m is the number of network nodes. Clearly, SBTPN exhibits lower computational complexity for mining “AND” relations in large-scale networks than the other two methods. This is because it reduces the number of candidate “AND” structures by merging redundant transitions, achieving this by analyzing the intersection relations among the input places of transitions pointing to the same effect variable.

VII. COMPARATIVE EXPERIMENT

An extension of the solar panel manufacturing process discussed in [42], which is based on the case introduced in Section II, is presented in [Fig. 3](#). It is used to validate the advantages of the proposed SBTPN over existing temporal causal discovery methods in temporal knowledge mining and RCA. In [Fig. 3](#), a place represented by a gray, yellow, or red circle indicates that a silicon rod, silicon wafer, or solar panel is

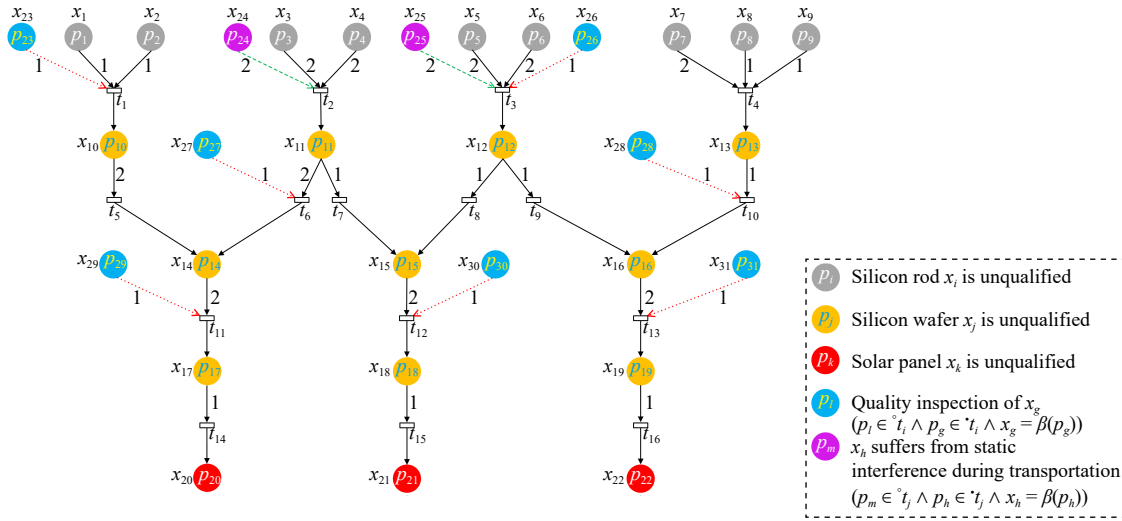


Fig. 3. Quality anomalies of solar panel manufacturing process.

unqualified, respectively. A place pictured by a blue circle represents quality inspection of a product. A place pictured by a purple circle represents that a product suffers from static interference during transportation. A transition reveals the causal mechanism between its input places-associated variables and its output places-associated variables. For a place represented by a blue (purple) circle and its associated transition, the variable corresponding to the place has an inhibitory (catalytic) effect on the causal rule corresponding to the transition.

The functions in Table I can be used to characterize the causal mechanism behind the case in Fig. 3. In Table I, u_i^τ ($i \in \{1, 2, \dots, 9, 23, 24, \dots, 31\}$) is a random number generated from the standard uniform distribution $U(0, 1)$ at the τ -th sampling point of variable x_i , and ε_j^τ ($j \in \{13, 20, 21, 22\}$) is a Gaussian-distributed white noise with mean 0 and standard deviation 0.005. The “AND” relations among the cause variables are depicted by the product of these variables. There are no strict restrictions on the generation mechanism of the “OR” relations among the cause variables, except that they cannot be characterized in the form of product of these variables. For transition t_k with $\bullet t_k = \bigcup_{a \in \mathbb{N}_h} \{p_a\}$ and $t_k^\bullet = p_j$, if it is affected by the catalytic effect of place p_x and the inhibitory effect of place p_y , the values of $\beta(p_x)$ and $\beta(p_y)$ are updated such that $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{p}_a\}, \hat{p}_{xj}) > \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{p}_a\})$ and $\Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{p}_a\}, \hat{p}_{yj}) < \Pr(\hat{p}_j | \bigcup_{a \in \mathbb{N}_h} \{\hat{p}_a\})$. Based on Table I, a dataset named Syn1000¹, which consists of 1000 data samples, is generated to verify the performance of the proposed SBTPN in mining complex temporal causal relations.

A. Performance in Temporal Knowledge Mining

This section evaluates the performance of the proposed SBTPN in temporal knowledge mining on the Syn1000 dataset. TCDF [39], PCMC [43], [44], MTE [34], and MLP-GC [45], which are four important temporal causality mining methods, receive much attention in the field of RCA. Considering their state-of-the-art performance in temporal causality mining, they are employed as SBTPN’s peers for temporal

TABLE I
CAUSAL FUNCTIONS CORRESPONDING TO THE CASE IN FIG. 3

Causal function		
$x_i^\tau = \begin{cases} u_i^\tau, & \tau = 1 \\ 0.7 \sin(1.3\pi x_i^{\tau-1}) + 0.3u_i^\tau, & \tau > 1 \end{cases}$	$\tau = 1$	$x_{16}^\tau = 0.5(x_{12}^{\tau-1} + 0.85x_{13}^{\tau-1} + 0.15(1 - x_{28}^{\tau-1}))$
$i = 1, 2, \dots, 9, 23, 24, \dots, 31$		
$x_{10}^\tau = 0.9x_1^{\tau-1}x_2^{\tau-1} + 0.1(1 - x_{23}^{\tau-1})$		$x_{17}^\tau = 0.8x_{14}^{\tau-2} + 0.1(1 - x_{29}^{\tau-1})$
$x_{11}^\tau = 0.9x_3^{\tau-2}x_4^{\tau-2} + 0.1x_{24}^{\tau-2}$		$x_{18}^\tau = 0.8x_{15}^{\tau-2} + 0.15(1 - x_{30}^{\tau-1})$
$x_{12}^\tau = 0.8x_5^{\tau-2}x_6^{\tau-2} + 0.1x_{25}^{\tau-2} + 0.1(1 - x_{26}^{\tau-1})$		$x_{19}^\tau = 0.85x_{16}^{\tau-2} + 0.1(1 - x_{31}^{\tau-1})$
$x_{13}^\tau = 0.95x_7^{\tau-2}x_8^{\tau-1}x_9^{\tau-1} + 0.05\varepsilon_{13}^\tau$		$x_{20}^\tau = 0.9x_{17}^{\tau-1} + 0.1\varepsilon_{20}^\tau$
$x_{14}^\tau = 0.5(x_{10}^{\tau-2} + 0.85x_{11}^{\tau-2} + 0.15(1 - x_{27}^{\tau-1}))$		$x_{21}^\tau = 0.85x_{18}^{\tau-1} + 0.15\varepsilon_{21}^\tau$
$x_{15}^\tau = 0.5(x_{11}^{\tau-1} + x_{12}^{\tau-1})$		$x_{22}^\tau = 0.8x_{19}^{\tau-1} + 0.2\varepsilon_{22}^\tau$

knowledge mining as well as RCA. Specifically, TCDF adopts dilated convolutions to discover temporal causality and is able to deal with hidden confounders. PCMC can detect and quantify causal relations in large nonlinear time series datasets. MTE utilizes transfer entropy to learn temporal causality. MLP-GC is a Granger causality analysis approach using multilayer perceptron.

The mining results of SBTPN and PCMC on the Syn1000 dataset, which are shown in Fig. 4, are used to illustrate the advantage of SBTPN in mining complex temporal causalities. From Fig. 4(a), SBTPN can discover the “AND” rule: IF p_7 AND p_8 AND p_9 THEN p_{13} , as well as the “OR” rule: IF p_{11} OR p_{12} THEN p_{15} , which are consistent with the ground truth shown in Fig. 3. From Fig. 4(b), PCMC fails to explicitly distinguish the difference between such “AND” and “OR” structures. In fact, all existing temporal causal discovery methods suffer from this issue. Moreover, as seen in Fig. 4(a), SBTPN can capture the inhibitory effect of place p_{31} on the causal rule: IF p_{16} THEN p_{19} , which is in line with the fact given in Fig. 3. From Fig. 4(b), PCMC treats p_{31} as the cause of p_{19} , which fails to depict the fact that quality inspection can prevent the circulation of defective products in manufacturing process. The inability to properly deal with synergy effects is

¹ <https://github.com/netlearninglabs/SBTPN>

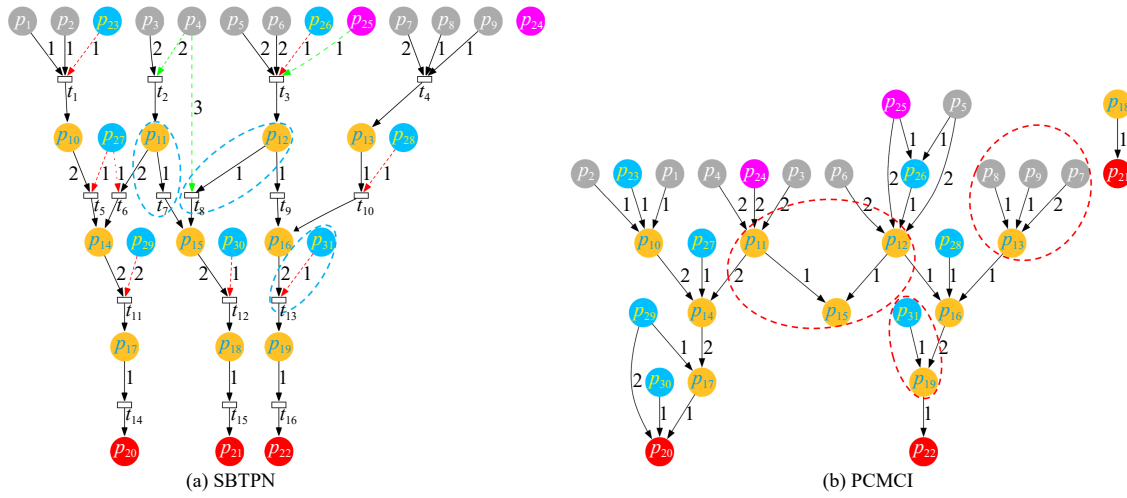


Fig. 4. Causal graphs discovered by SBTPN and PCMCi.

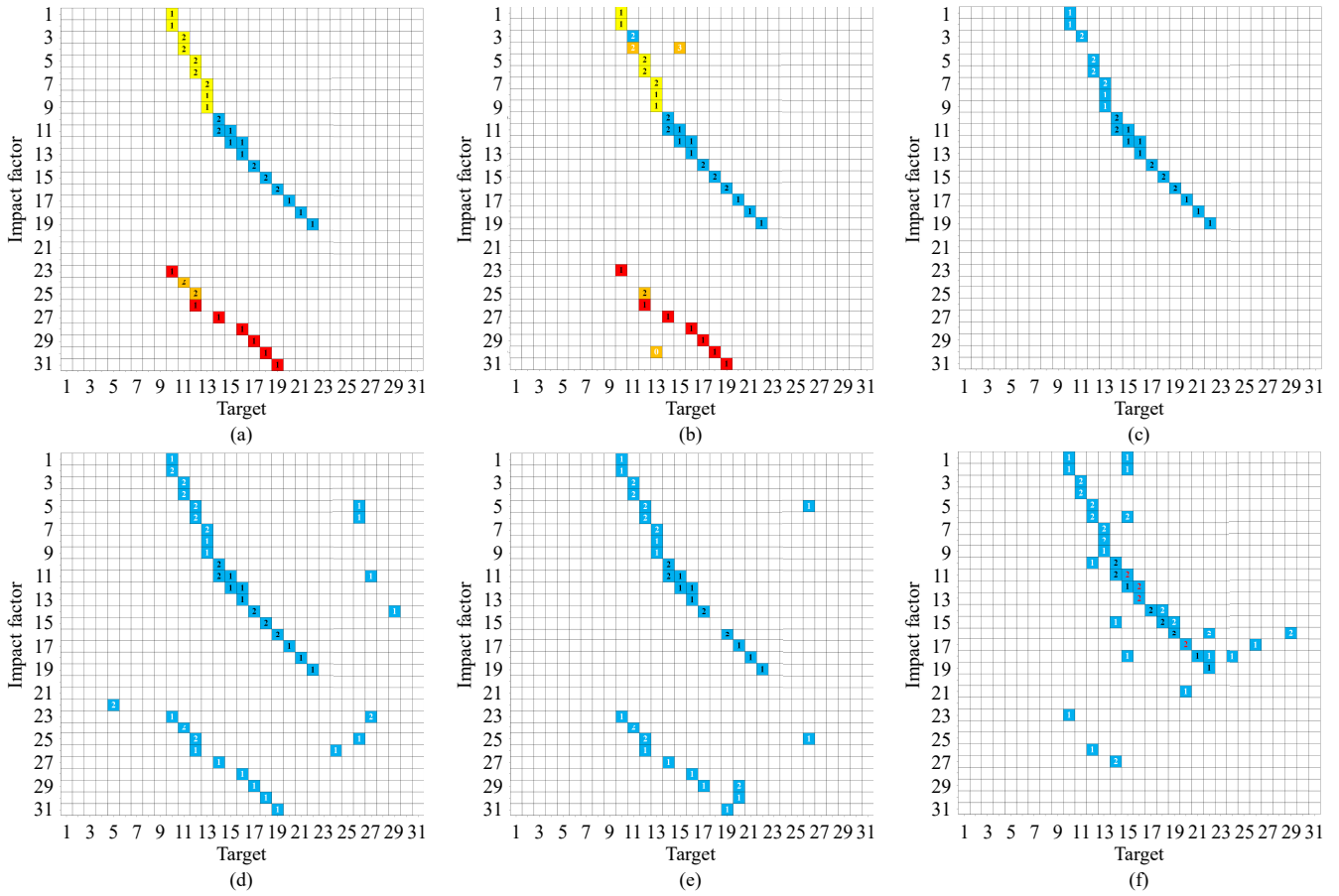


Fig. 5. Causality matrices generated by all compared methods ((a) Ground truth; (b) SBTPN; (c) TCDF; (d) MTE; (e) PCMCi; (f) MLP-GC).

another limitation of existing temporal causal discovery methods.

Fig. 5 shows the performances of all compared methods in discovering complex temporal causalities on the Syn1000 dataset. In Fig. 5, the yellow (blue) squares in the same column represent the “AND” (“OR”) relations among the impact factors pointing to the target variable. A blue square in a column without a peer represents a simple causal relation between an impact factor and its target variable. A red

(orange) square means the inhibitory (catalytic) effect of an impact factor on its target variable. The number in a square denotes the time delay of an impact factor affecting its target variable. A black (red) number denotes a correct (incorrect) time delay captured by a causal discovery model. A white number means a discovered time delay that corresponds to an incorrect causal relation or synergy effect. Fig. 5(a) represents the true causal relations shown in Fig. 3. For the “OR” and “AND” structures pointing to the same effect variable, the

TABLE II
PERFORMANCE IN TEMPORAL KNOWLEDGE MINING

Model	$Recall_d$	$Precision_d$	FI_d	$Recall_a$	$Precision_a$	FI_a	$Recall_s$	$Precision_s$	FI_s	$Recall_e$	$Precision_e$	FI_e
SBTPN	0.95	1.00	0.97	0.75	1.00	0.86	0.89	0.73	0.80	0.87	0.84	0.85
TCDF [39]	0.95	1.00	0.97	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	0.67	1.00	0.80
MTE [34]	0.90	0.54	0.68	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	0.63	0.54	0.58
PCMCi [43]	0.90	0.63	0.74	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	0.63	0.06	0.63
MLP-GC [45]	1.00	0.55	0.71	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	0.53	0.42	0.47

“OR” structure can weaken the correlation between the cause variables in the “AND” structure and the effect variable, thus increasing the difficulty of identifying the “AND” structure. In addition, the stronger the synergy effect of a synergy variable on an effect one, the higher the correlation between them. This increases the possibility that the synergy effect is mistakenly identified as the causal relation between them. As shown in Fig. 5(b), despite facing these challenges, SBTPN can correctly discover most of the “AND/OR” relations and synergy effects shown in Fig. 5(a), and the number of incorrect causal relations and synergy effects identified by it is small. Since SBTPN is presented based on TCDF, Fig. 5(c) can be obtained by neglecting the “AND/OR” relations and synergy effects in Fig. 5(b). From Figs. 5(c)–5(f), MTE and PCMCi identify most of the synergy effects as the causal relations, whereas TCDF (MLP-GC) neglects all (most) of the synergy effects. These two ways of dealing with synergy effects can weaken the reliability of RCA, to be discussed in the next section.

Next, the performance of the compared models in mining complex temporal relations is quantitatively analyzed using the following metrics:

1) $Recall_d$: Proportion of correctly detected direct causal relations to the total direct causal relations in the true causal network;

2) $Precision_d$: Proportion of correctly detected direct causal relations to the total direct causal relations in the generated causal network;

3) FI_d : The harmonic average of $Recall_d$ and $Precision_d$;

4) $Recall_a$: Proportion of correctly detected “AND” relations to the total “AND” relations in the true causal network;

5) $Precision_a$: Proportion of correctly detected “AND” relations to the total “AND” relations in the generated causal network;

6) FI_a : The harmonic average of $Recall_a$ and $Precision_a$;

7) $Recall_s$: Proportion of correctly detected synergy effects to the total synergy effects in the true causal network;

8) $Precision_s$: Proportion of correctly detected synergy effects to the total synergy effects in the generated causal network;

9) FI_s : The harmonic average of $Recall_s$ and $Precision_s$;

10) $Recall_e$: Proportion of correctly detected time delays to the total time delays in the true causal network;

11) $Precision_e$: Proportion of correctly detected time delays to the total time delays in the generated causal network; and

12) FI_e : The harmonic average of $Recall_e$ and $Precision_e$.

These metrics are adopted for the following reasons: They

provide a comprehensive characterization of causal discovery models’ ability to identify “AND/OR” relations, synergy effects, and the time delays associated with their generation.

Table II quantitatively analyzes the performance of all compared models in temporal knowledge mining. The bold fonts represent the best results for each column. From Table II, SBTPN is the best method for mining direct causal relations, AND relations, synergy effects, and the time delays associated with causal or synergy effects among the five compared ones. The performance of SBTPN in knowledge mining and RCA is sensitive to its parameters, especially threshold η_1 used to identify catalytic effects and threshold η_2 used to determine inhibitory effects. A detailed sensitivity analysis of these two parameters is provided in the Supplementary File.

B. Performance in RCA

Based on Table I, 500 process inspection records named Syn500² are randomly generated to evaluate the performance of the compared methods in RCA. This study focuses on the abnormalities of the end products x_{20} , x_{21} , and x_{22} . The Supplementary File provides the principles of RCA based on the compared methods. According to the causal relations shown in Fig. 5, the true root causes of these three variables and their root causes predicted by all compared methods can be derived.

The true root causes of an anomalous variable x_k $k \in \{20, 21, 22\}$ on the sample s of Syn500 can be expressed as a vector $u_k^s = [u_1, u_2, \dots, u_{31}]$, where $u_i = 1$ ($u_i = 0$) represents that x_i is (is not) the true root cause of x_k , $i \in \{1, 2, \dots, 31\}$. The root causes of x_k derived from a causal discovery model M_v on s can also be represented by a vector $v_k^s = [v_1, v_2, \dots, v_{31}]$, where $v_i = 1$ ($v_i = 0$) indicates that x_i is (is not) the root cause of x_k . Then, the performance of M_v in predicting the root causes of x_k on s is defined as

$$\cos(u_k^s, v_k^s) = \frac{\sum_{i=1}^{31} u_i v_i}{\sqrt{\sum_{i=1}^{31} u_i^2} \sqrt{\sum_{i=1}^{31} v_i^2}}. \quad (9)$$

The larger the value of $\cos(u_k^s, v_k^s)$, the better the performance of M_v in predicting the root causes of x_k on s . On this basis, the RCA accuracy of M_v on sample set S can be expressed as $(\sum_{s \in S} \cos(u_k^s, v_k^s))/|S|$, where $|S|$ represents the size of S . Table III shows the performance of all compared methods in RCA for x_{20} , x_{21} , and x_{22} on Syn500.

From Table III, in terms of RCA for x_{20} , x_{21} , and x_{22} ,

² <https://github.com/netlearninglabs/SBTPN>

TABLE III
PERFORMANCE IN RCA FOR x_{20} , x_{21} , AND x_{22}

Model	RCA accuracy		
	x_{20}	x_{21}	x_{22}
SBTPN	0.97	0.97	1.00
TCDF	0.74	0.83	0.89
MTE	0.74	0.84	0.88
PCMCi	0.24	0.83	0.88
MLP-GC	0.35	0.75	0.78

SBTPN outperforms its four peers in RCA accuracy by 23% to 73%, 13% to 22%, and 11% to 22%, respectively, which is highly impressive. Clearly, SBTPN performs the best in RCA for all the three end products among the five peers. This is because its four peers fail to consider the role of inhibitory effects in blocking causal effects. Consider the causal structure in Fig. 6(a). Suppose that x_{20}^{τ} , $x_{17}^{\tau-1}$, and $x_{14}^{\tau-3}$ are all unqualified, and $x_{14}^{\tau-3}$ is quality checked at time $\tau-2$ (i.e., $x_{29}^{\tau-2}$ is true). Due to the fact that quality inspection can prevent the circulation of defective products between upstream and downstream processing, $x_{14}^{\tau-3}$ being unqualified is not the cause of x_{20}^{τ} being unqualified. In this case, the cause of x_{20}^{τ} being unqualified is $x_{17}^{\tau-1}$ being unqualified. As shown in Figs. 6(c)–6(f), the four competing models are unable to depict the inhibitory effect of p_{29} on the causal relation between p_{14} and p_{17} . It can be inferred from their mining results that x_{14} being unqualified is the cause of x_{20} being unqualified, which is inconsistent with the fact. The inability to properly depict synergy effects is the reason why the four competing models perform worse than SBTPN in RCA for x_{20} . In addition, as seen in Table III, PCMCi and MLP-GC perform worse than the other three models in RCA for x_{20} . From Fig. 6, this is because the quality of the local causal structures pointing to x_{20} mined by PCMCi and MLP-GC is inferior to the quality of those discovered by the other three models. Usually, the higher the quality of the discovered causal structures, the higher the credibility of RCA.

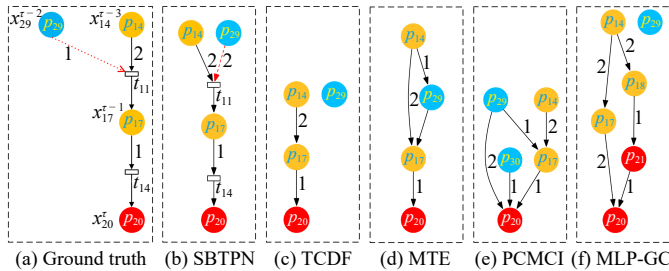


Fig. 6. Local causal structures pointing to x_{20} .

VIII. EXTENDED APPLICATIONS OF SBTPN

The Tennessee-Eastman (TE) process [46], a classical process industry simulation platform, has attracted widespread attention from researchers in the field of RCA [35], [47]. It is used to further validate the generalizability of SBTPN. As shown in Fig. 7, it consists of five main units: a reactor, a con-

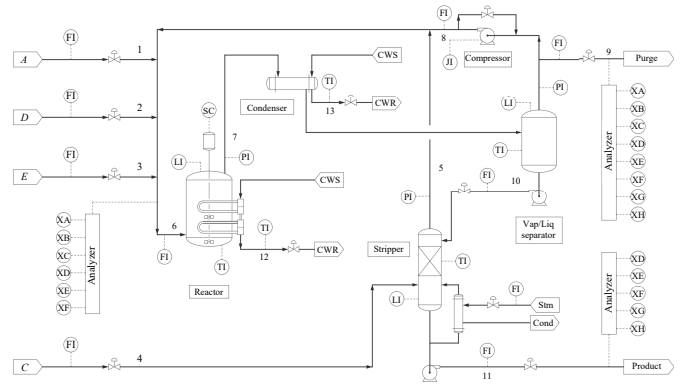
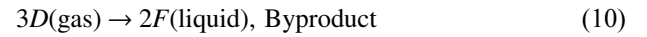
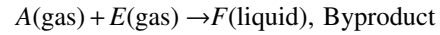
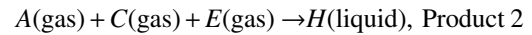
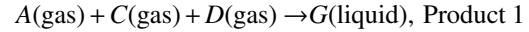


Fig. 7. Process flow diagram of the TE process.

denser, a vapor-liquid separator, a compressor, and a stripper. Four gas reactants A , D , E , and C are used to produce two liquid products G and H and one by-product F in the presence of an inert component B . The involved reaction equations are as follows:



There are 12 process manipulated variables named XMV(1)–(12), 22 continuous process measurement variables named XMEAS(1)–(22), and 19 sampled process measurement variables named XMEAS(23)–(41) in the process. 20 typical faults named IDV(1)–(20) are simulated in the process. 960 observed samples are generated to simulate each fault, and the disturbance is added after the 160th sample. The sampling interval is 3 min. The detailed description of all variables and faults can be found in [46]. Fault case IDV(1) is used to verify the performance of the proposed SBTPN in knowledge mining and RCA toward alarm variables. The continuous process measurement variables XMEAS(1)–(22) and process manipulated variables XMV(1)–(12) are employed as the candidate alarm variables.

A. Performance in Knowledge Mining

Any candidate alarm variable u_i can be decomposed into a high-level alarm variable $u_i(H)$ and a low-level one $u_i(L)$ by using the 3σ principle [48]. The variable whose alarm probability does not change substantially under normal conditions and IDV(1) is not considered in knowledge mining and RCA. Then, all alarm variables under IDV(1) can be obtained, as shown in Table IV. In the following, to save space, XMEAS(i) and XMV(j) are abbreviated as x_i ($i = 1, 2, \dots, 22$) and v_j ($j = 1, 2, \dots, 12$), respectively.

To identify the root cause of the alarms in Table IV, the causal relations among them should be determined first. SBTPN is employed to learn the complex causal relations among the alarms. The discovered complex causal relations are shown in Fig. 8. Note that: 1) Some variables in Table IV have no causal relations or synergy effects with other vari-

TABLE IV
ALARM VARIABLES UNDER IDV(1)

Variable	Description	Variable	Description	Variable	Description
$x_1(H)$	The flow rate of A feed in stream 1 is high	$x_{10}(H)$	The purge rate of stream 9 is high	$x_{19}(H)$	The flow rate of stripper steam is high
$x_3(L)$	The flow rate of E feed in stream 3 is low	$x_{11}(L)$	Separator temperature is low	$x_{20}(L)$	Compressor power is low
$x_4(L)$	The flow rate of A and C feed in stream 4 is low	$x_{11}(H)$	Separator temperature is high	$x_{20}(H)$	Compressor power is high
$x_7(L)$	Reactor pressure is low	$x_{13}(L)$	Separator pressure is low	$x_{21}(H)$	Reactor water temperature is high
$x_7(H)$	Reactor pressure is high	$x_{13}(H)$	Separator pressure is high	$x_{22}(L)$	Separator water temperature is low
$x_8(L)$	Reactor level is low	$x_{16}(L)$	Stripper pressure is low	$v_9(H)$	The opening degree of the stripper steam valve is high
$x_8(H)$	Reactor level is high	$x_{16}(H)$	Stripper pressure is high		
$x_{10}(L)$	The purge rate of stream 9 is low	$x_{18}(H)$	Stripper temperature is high		

TABLE V
CAUSAL RULES IN FIG. 8 WITH THEIR TERMINAL VARIABLES AND SYNERGY FACTORS

Causal rule	Terminal variable	Catalytic factor	Inhibitory factor	Causal rule	Terminal variable	Catalytic factor	Inhibitory factor
$x_4(L) \rightarrow x_1(H)$	$x_1(H)$	—	—	$x_7(L) \rightarrow x_{16}(L)$	$x_{16}(L)$	$x_{20}(H) 0.05$ $x_{10}(H) 0.12$	$x_8(H) 0.09$ $x_{16}(H) 0.01$
$x_4(L) \& v_9(H) \rightarrow x_{18}(H)$	$x_{18}(H)$	—	—	$x_7(L) \rightarrow x_8(L)$	$x_8(L)$	$x_3(L) 0.63$ $x_{11}(H) 0.53$ $x_{20}(H) 0.33$ $x_{10}(H) 0.26$	$x_8(H) 0.00$ $x_{10}(L) 0.03$ $x_{16}(H) 0.09$
$v_9(H) \rightarrow x_{19}(H)$	$x_{21}(H)$	—	—	$x_{22}(L) \rightarrow x_{11}(L)$	$x_{13}(L)$	$x_7(H) 0.00$ $x_8(H) 0.22$ $x_{13}(H) 0.00$ $x_{16}(H) 0.06$ $x_{20}(H) 0.00$	$x_3(L) 0.53$ $x_{11}(H) 0.69$ $x_{10}(H) 0.09$
$x_{19}(H) \rightarrow x_{21}(H)$	$x_{21}(H)$	$x_8(H) 0.46$ $x_{11}(L) 0.32$ $x_{13}(L) 0.23$ $x_{16}(L) 0.36$ $x_{20}(L) 0.54$ $x_{22}(L) 0.46$	$x_8(L) 0.03$ $x_{20}(H) 0.10$	$x_{22}(L) \rightarrow x_{11}(L)$	$x_8(L)$	$x_7(H) 0.13$ $x_8(H) 0.04$ $x_{13}(H) 0.14$ $x_{16}(H) 0.09$ $x_{20}(H) 0.28$	$x_3(L) 0.59$ $x_{11}(H) 0.47$ $x_{10}(H) 0.23$
$x_{11}(H) \rightarrow x_{16}(H)$	$x_{16}(H)$	—	$x_7(L) 0.00$ $x_{13}(L) 0.00$	$x_{11}(L) \rightarrow x_8(L)$	$x_8(L)$	$x_3(L) 0.61$ $x_{10}(H) 0.26$	—
$x_7(H) \rightarrow x_{16}(H)$	$x_{16}(H)$	$x_{20}(L) 0.24$	$x_8(L) 0.11$ $x_{10}(H) 0.08$	$x_{11}(L) \rightarrow x_{13}(L)$	$x_{13}(L)$	$x_8(H) 0.14$	$x_{20}(H) 0.00$

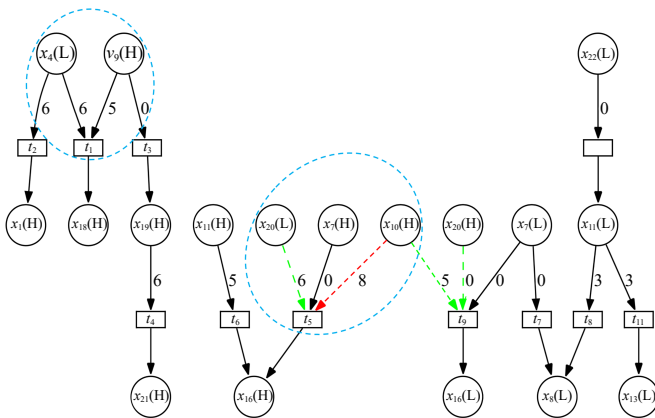


Fig. 8. The discovered complex causal relations among alarm variables.

ables. Therefore, they are not depicted in Fig. 8; and 2) To present synergy effects in an intuitive and concise manner, Fig. 8 only shows two typical synergy effects mined by SBTPN. All the discovered synergy effects are detailed in Table V.

The opening degree of the stripper steam valve determines

the flow rate of stripper steam. If the opening degree of the stripper steam valve is high (i.e., $v_9(H)$ is true), the flow rate of stripper steam is high (i.e., $x_{19}(H)$ is true). From Fig. 8, SBTPN can discover the causal relation between $v_9(H)$ and $x_{19}(H)$. Removing impurities from the products in the stripper is an endothermic process. The temperature of the stripper is determined by the flow rate of the stripper steam as well as the supply of the reactants in the stripper. The higher its flow rate and the lower supply of the reactants in the stripper, the higher temperature of the stripper. In fact, both $v_9(H)$ being true (i.e., the opening degree of the stripper steam valve is high) and $x_4(L)$ being true (i.e., the flow rate of A and C feed in stream 4 is low) are the necessary conditions for $x_{18}(H)$ being true (i.e., stripper temperature is high). That is, there is an “AND” relation between $v_9(H)$ and $x_4(L)$ that points to $x_{18}(H)$. From Fig. 8, SBTPN can capture this “AND” relation represented by transition t_1 . However, existing causal discovery methods, e.g., Granger causality-based [9], [22] and deep learning-based ones [10], [23], are unable to learn such relation from MTS explicitly.

In fact, there are synergy effects of non-causal variables on causal rules in the alarm network in Fig. 8. Consider the following cases. The pressure of the stripper is closely related to

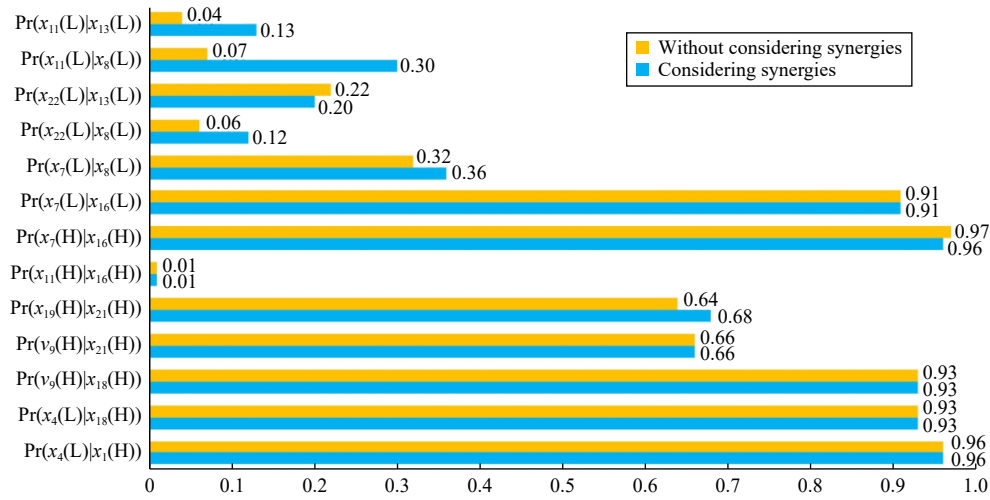


Fig. 9. Root cause score of a non-terminal variable w.r.t. a terminal one.

that of the reactor. An increase (decrease) in reactor pressure can lead to an increase (decrease) in stripper pressure. From Fig. 8, SBTPN can mine the causal relation between $x_7(H)$ (Reactor pressure is high) and $x_{16}(H)$ (Stripper pressure is high). Actually, “the compressor power being low” can increase the possibility of “the stripper pressure being high” caused by “the reactor pressure being high”. The reasons are as follows. A low compressor power can reduce the recovery rate of the reactants in the separator. This means that the pressure of the separator is high, which implies a high pressure value of the stripper. Thus, $x_{20}(L)$ (Compressor power is low) has a catalytic effect on rule R_1 : IF “ $x_7(H)$ is true” THEN “ $x_{16}(H)$ is true”. Contrarily, “the purge rate of stream 9 being high” can reduce the possibility of “the stripper pressure being high” caused by “the reactor pressure being high”. This is because a high purge rate of stream 9 can lead to a low pressure of the separator, resulting in a low pressure of the stripper. Therefore, $x_{10}(H)$ (The purge rate of stream 9 is high) has an inhibitory effect on R_1 . From Fig. 8, SBTPN can discover these catalytic/inhibitory effects from MTS. However, existing causal discovery methods, e.g., Granger causality-based [9], [22] and deep learning-based ones [10], [23], do not focus on learning such effects from MTS.

B. Performance in RCA

IDV(1) causes a step change in the A/C feed ratio in stream 4, resulting in a decrease in the amount of A in recycling stream 5. This disturbance is detected by the component analyzer associated with stream 6. Then, the feedback control system attempts to reduce the disturbance by increasing the flow rate of A feed in stream 1. This study focuses on the root causes of high/low alarm variables. According to [49], $x_4(L)$ (The flow rate of A and C feed in stream 4 is low) is the root cause of the alarms under IDV(1).

An alarm root cause variable should be able to cause alarms for other variables. Therefore, all variables corresponding to the non-terminal nodes in Fig. 8 are considered as the potential root causes. In this paper, the variables corresponding to terminal nodes are referred to as terminal variables. For a terminal variable x_j with its direct/indirect cause x_i , the probability of $x_i^{\tau-e_{ij}}$ being true conditioned on x_j^{τ} being true is uti-

lized to measure whether the abnormality of x_i is a root cause of the abnormality of x_j , where e_{ij} represents the time delay of the direct/indirect causal relation between x_i and x_j . Table V presents all the causal rules in Fig. 8 along with their corresponding terminal variables and synergy factors. In Table V, for the terminal variable x_j and synergy factor x_h associated with the same causal rule, the number corresponding to x_h indicates the probability of $x_h^{\tau-e_{hj}}$ being true conditioned on x_j^{τ} being true, where e_{hj} represents the time delay of x_h affecting x_j .

For each non-terminal variable x_i in Fig. 8 with its corresponding terminal one x_j , the possibility that x_i is the root cause of x_j is evaluated by using the proposed RCA method and Table V, yielding Fig. 9.

From Fig. 9, the root cause scores of non-terminal variables $x_4(L)$ and $x_7(H)$ are higher than those of other non-terminal variables. According to Table V, the causal rules associated with $x_4(L)$ are not affected by any catalytic/inhibitory effects. Therefore, the root cause score of $x_4(L)$ remains unchanged regardless of whether synergy effects are considered or not. However, the causal rule associated with $x_7(H)$ is more affected by inhibitory effects than by catalytic effects. Therefore, the root cause score of $x_7(H)$ when considering synergy effects is lower than that when not considering synergy effects. Theoretically, the causal relation between an alarm root cause variable and its effect variable cannot be blocked by inhibitory effects. Otherwise, the abnormal state of the alarm root cause variable cannot be transmitted to its effect variable. That is, an alarm root cause variable should be insensitive to inhibitory effects. Therefore, $x_4(L)$ is more likely than $x_7(H)$ to be the root cause of the alarms. In Table V, for the causal rule “ $R_1: x_{22}(L) \rightarrow x_{11}(L)$ ” pointing to the terminal variable $x_{13}(L)$, although the number of its catalytic factors is greater than that of its inhibitory factors, the possibility of its inhibitory factors being present is greater than that of its catalytic factors being present under the condition that $x_{13}(L)$ is in an alarm state. From Fig. 9, this makes the root cause score of $x_{22}(L)$ with respect to $x_{13}(L)$ lower when synergy effects are considered than when they are not. In Table V, for the causal rule “ $R_2: x_{11}(L) \rightarrow x_{13}(L)$ ” pointing to the terminal variable $x_{13}(L)$, the possibility of its catalytic factors being present is

greater than that of its inhibitory factors being present under the condition that $x_{13}(L)$ is in an alarm state. From Fig. 9, this makes the root cause score of $x_{11}(L)$ with respect to $x_{13}(L)$ higher when synergy effects are considered than when they are not. From the above analysis, synergy effects play an important role in RCA. Existing RCA methods [10], [22], [23] do not consider such effects. The proposed SBTPN can capture the impact of synergy effects on RCA.

IX. CONCLUSIONS

By considering “AND/OR” relations and synergy effects, this study proposes a new knowledge mining and RCA methodology for MTS called SBTPN. TCDF [39] is employed to learn simple temporal causal relations from MTS. Based on TCDF, the concept of dependence strength index is defined to quantify the impact of an input time series on a target one. By using the dependence strength index, this study discovers the “AND/OR” relations among causes as well as the synergy effects owned by non-causal variables on causal rules from MTS. PNs and BNs are fused to model the “AND/OR” relations and synergy effects toward MTS, resulting in an SBTPN. Based on the discovered SBTPN as well as the dependence strength index, this study presents an RCA method considering “AND/OR” relations and synergy effects. A case of solar panel manufacturing process and the TE process are adopted to evaluate the performance of SBTPN in temporal knowledge mining and RCA. The experimental results in temporal knowledge mining show that SBTPN can effectively discover the complex temporal causalities containing “AND/OR” relations and synergy effects. The experimental results in RCA indicate that considering such relations and effects can enhance the credibility of RCA.

Improving the computational efficiency of SBTPN on large-scale MTS is a key focus of our future work. Graph neural networks (e.g., GraphSAGE [50] and dynamic graph neural networks [51]) and neural network-based causal discovery models (e.g., neural causal models [52]) are capable of efficiently modeling the dependencies among nodes. Nevertheless, they do not explicitly capture “AND/OR” relations, synergy effects, and the time delays inherent in these factors. Our next work plans to integrate these models’ efficiency in graph structure learning with SBTPN’s strengths in fine-grained causal relations mining, thereby enhancing the effectiveness and efficiency of SBTPN in handling large-scale systems [53]–[56].

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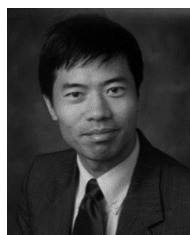
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